

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460722.6519 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.019
Transit depth (Rp/Rs)^2	2.11 +/- 0.56 [%]
Orbital Inclination (inc)	84.19 +/- 1.43
Airmass coefficient 1 (a1)	0.6112 +/- 0.0047
Airmass coefficient 2 (a2)	0.353 +/- 0.079
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	#3 - [546, 391]
Optimal Aperture	3.9
Optimal Annulus	9.87
Transit Duration (day)	0.039 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.019 vs 0.11 (deviation 1.75σ)
Duration fit vs published	0.039 d vs 0.1208 d (deviation 3.38σ)
Mid-time O–C	-26.9 min
Depth SNR	7.2
Residual scatter	0.76 %

Light curve

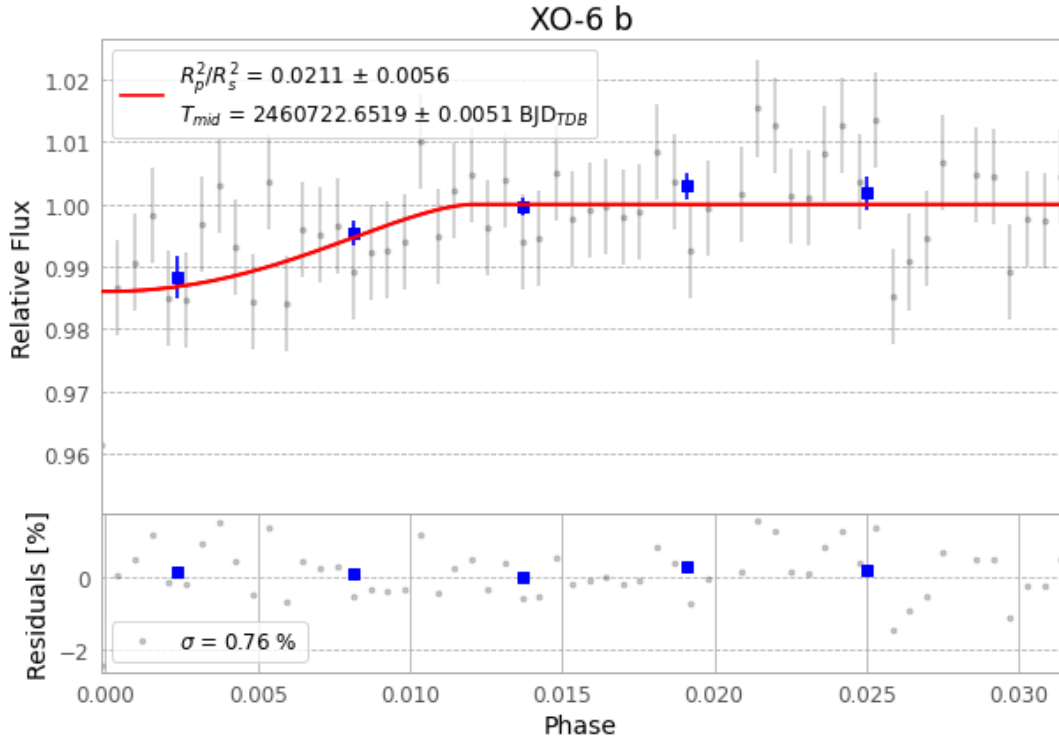


Figure 1 — FinalLightCurve_XO-6 b_2025-02-15.png (SHA-256 ab00a5d66bf822d6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [414, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ec242d8978f1730c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

58 FITS frames (38 MB), XO-6250216033315.FITS → XO-6250216062416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-250216.log (086ed89f93e2...), FinalLightCurve_XO-6 b_2025-02-15.png (ab00a5d66bf8...), FinalLightCurve_XO-6 b_2025-02-15.csv (fe0ae430286a...)

Compute resources

Wall time 565.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:03Z.